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Genesis of the main types of sulphide ore in the Jinchuan Ni–Cu–PGE deposit, NW China: Constraints from texture and mineral chemistry of pyrrhotite

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The Jinchuan Ni–Cu–PGE deposit is one of the three largest magmatic Ni deposits in the world. Disseminated, net-textured, Cu-rich sulphide ores and massive sulphide ores are the most important ore types. The genesis of different ore types and the mineralization processes of the Jinchuan deposit remain controversial. Three hypotheses have been proposed by previous studies, including the in situ sulphide segregation of sulphide-bearing magma, deep segregation-multiple injections, and hydrothermal superimposition. To understand the genesis of different ore types, we studied the occurrence, intergrowth textures, and mineral chemistry of pyrrhotite in four types of sulphide ore using magnetic colloid experiment and electron probe microanalysis. Pure hexagonal pyrrhotites (Hpo) in disseminated and net-textured sulphide ores display higher M/S values, indicating a slow cooling process in a high-temperature and low-S magmatic system. Irregular intergrowth pyrrhotites occur in disseminated, net-textured, and Cu-rich sulphide ores. Dusty, stellate, and patchy monoclinic pyrrhotites (Mpo) occur along the fissures and grain boundaries of Hpo or the contact boundaries of silicate minerals, suggesting the interaction of Hpo and sulphur- and/or oxygen-rich hydrothermal fluids. Monoclinic pyrrhotite occurs as roughly parallel lamellae in the matrix of Hpo and lacks of pyrite porphyroblasts in massive sulphide ores, indicating a rapid reduction of temperature in a high-S magmatic system. The Cu-rich sulphide ores are characterized by a relatively high proportion of chalcopyrite and comparatively high Pt and Pd (PPGE) and low Os, Ir, Ru (IPGE), and Rh concentrations, indicating a fractional crystallization of monosulphide solid solution from sulphide melts, and then the residual sulphide melts mineralized in a local place.

KEYWORDS

hexagonal pyrrhotite (Hpo), Jinchuan Ni–Cu–PGE deposit, monoclinic pyrrhotite (Mpo), monosulphide solid solution (MSS), ore types

1 | INTRODUCTION

The Jinchuan Ni–Cu–PGE sulphide deposit is one of the three largest magmatic sulphide deposits in the world and the largest one in China. It contains more than 500 million metric tons of ores with 1.1 wt.% Ni, 0.7 wt.% Cu, 0.03 wt.% Co, and 0.66 ppm PGE (Tang & Li, 1995; Tang, Song, & Su, 2009). Previous studies of the Jinchuan deposit mainly focused on the most fundamental research questions, such as the composition and crystallization process of the parental magma, the segregation of immiscible sulphide liquid, and the collection and concentration of the metal-rich sulphide liquid (Barnes & Tang, 1999; Chai & Naldrett, 1992; Chen et al., 2008, 2009, 2013; De Waal, Xu, & Li, 2004; Duan et al., 2016; Lehmann et al., 2007;

Li & Ripley, 2011; Li, Xu, De Waal, Ripley, & Maier, 2004; Naldrett, 1999; Ripley, Sarkar, & Li, 2005; Song et al., 2012; Tang, 1996, 2002; Tang, Qian, & Jiang, 2006).

The genesis of sulphide ores in the Jinchuan deposit could be attributed to three different processes, including the in situ sulphide segregation of sulphide-bearing magma, deep segregation-multiple injections, and hydrothermal superimposition (Tang et al., 2009). However, there is little mineralogical evidence to propose a mechanism of ore-forming and physicochemical condition of mineralization until now. Pyrrhotite, one of the most important sulphide minerals, can occur as different phases depending on the range of crystallization temperature, oxygen fugacity, high sulphur or low-sulphur system in magmatic deposit, which differs from those formed by metamorphism and annealing in massive sulphide deposits

(Gu & Vokes, 1996; Gu et al., 1995; Yang et al., 2014; Zhou, Gu, & Li, 1998), or skarn copper deposits (Guo, Lu, Zhang, & Xu, 2010; Yang, Du, Cao, Zhang, & Liu, 2012). The textures and chemistry of pyrrhotite can indicate the genesis and mineralization process of ore deposit. Nakazawa and Morimoto (1970, 1971) sorted the numerous superstructures of pyrrhotite into five categories according to their multiplicities of the NiAs sublattice, including 2C, 4C, NC, MC, and NA. Becker, Villiers, and Bradshaw (2010) studied the mineralogy and crystallography of pyrrhotite from Ni-PGE ore deposits including Merensky reef, Nkomati, Phoenix, and Sudbury and sorted the structures of pyrrhotite into 6C intergrown with 2C, and 4C intergrown with 5C, pure 5C, and 4C. Naldrett, Singh, Krstic, and Li (2000) sorted the structures of pyrrhotite into NC intergrown with 2C, and pure NC in Voisey's Bay Cu-Ni sulphide deposit.

In this paper, we focus on the occurrence, intergrowth structures and analyse the mineral chemistry of pyrrhotite in the disseminated, net-textured, Cu-rich sulphide ores, and massive sulphide ores, to discuss the mineral crystallization environment and ore genesis of the Jinchuan deposit.

2 | GEOLOGICAL BACKGROUND

The Jinchuan Ni-Cu-PGE deposit is located in the Longshoushan terrane in the western portion of the North China Craton (Figure 1). The basement of the Longshoushan terrane is dominated by the Paleoproterozoic Longshoushan Group, which is composed of Baijiazui and Tamazigou

formations. The Baijiazui Formation is composed of migmatites, gneisses and marbles, whereas the Tamazigou Formation consists of schists and banded marbles (Li et al., 2005). The cover is composed of Mesoproterozoic and Neoproterozoic metasedimentary rocks. During the Neoproterozoic (~825 Ma), the Jinchuan intrusion intruded into the Paleoproterozoic Baijiazui Formation. In addition, abundant Palaeozoic granites occur in the Longshoushan terrane and the adjacent North Qilian Orogenic Belt to the south. The collision between North Qilian and southern Alxa took place in the Palaeozoic, indicated by ~450-Ma eclogites in the suture zone (Song, Niu, Su, & Xia, 2013).

The current outcrop area of the Jinchuan intrusion is ~6,500 m in length and 20 to 530 m in width, with downward extension reaching >1,000 m from the surface. The outcrop of the Jinchuan intrusion is lens-shaped, with an area of 1.34 km². The NW-trending intrusion dips to the southwest at oblique angles varying from 50° to 80°. The Jinchuan intrusion can be divided into several segments by a series of northeast-trending strike-slip faults, which were numbered as III, I, II, and IV from west to east, respectively (Figure 2a).

Segment I is located between faults F₈ and F₁₆ with 1,500 m in length and 30 to 320 m in width. The maximum width of the western part reaches 320 m, narrowing eastwards with a downward extension of up to 700 m. The slope trend of Segment I is 50°–60°, and it dips to the southwest. The sulphide ore body (No. 24) occurs in the middle and lower parts of Segment I. In recent years, a concealed ore body, named the “concealed Cu-rich ore body,” was found to the right of the No. 24 ore body in Segment I by on-going underground mining

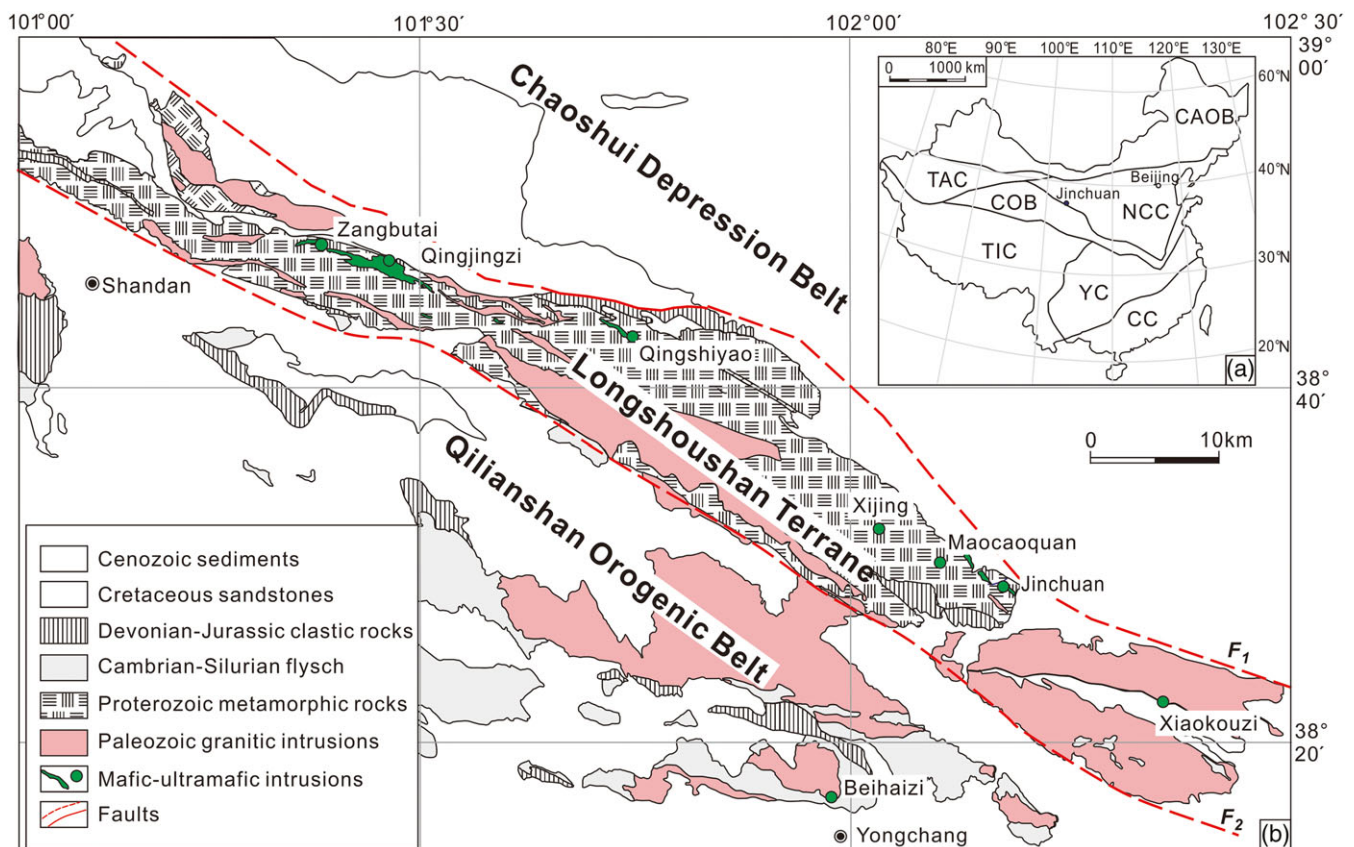


FIGURE 1 Simplified regional geological map of the Longshoushan terrane (modified after Song et al., 2012) Abbreviations: CAOB, Central Asian Orogenic Belt; CC, Cathaysia Craton; COB, Central Orogenic Belt; NCC, North China Craton; TAC, Tarim Craton; TIC, Tibet Craton; YC, Yangtze Craton.

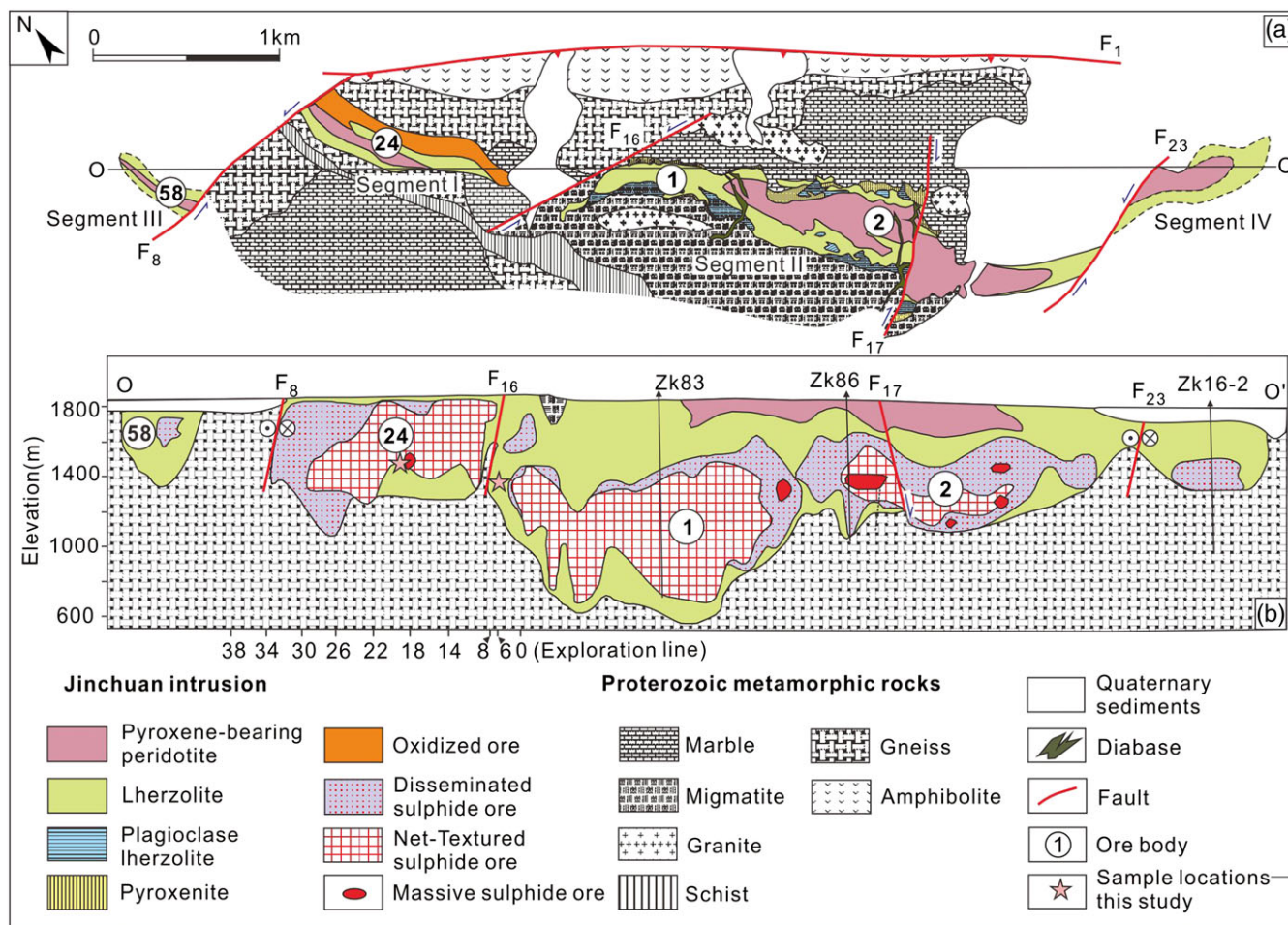


FIGURE 2 Geologic map (a) and longitudinal section (b) of the Jinchuan Ni-Cu-PGE deposit (modified after Song et al., 2012; Tang & Li, 1995), showing locations of major faults, intrusion segments, and ore bodies 58 (in Segment III), 24 (in Segment I), 1 (NW and central parts of Segment II), and 2 (SE part of Segment II), and Segment IV

operations. The Cu-rich ore body is located between Lines 6 and 8 (Figure 2b), with an elevation of 1,400 to 1,700 m and a footwall of F_{16} . The lens-shaped Cu-rich ore body dips to the southwest at 70° , with a maximum thickness of 40 m.

Segment II between faults F_{16} and F_{23} is the largest block of the Jinchuan intrusion, with >3,000 m in length and up to 500 m in width. It gradually widens from west to east with a maximum width of approximately 530 m and dips to the southwest with slopes of 50° to 80° with an overall strike of NW 50° . Along the exploration Line 26, Segment II can be divided into eastern and western mining areas, which are labelled as the No. 1 and No. 2 ore bodies, respectively. The No. 1 ore body is tabular in shape and constitutes almost the entire amount of igneous rocks at depth. It is primarily composed of net-textured sulphide ore, as well as minor disseminated and massive sulphide ores. The smaller No. 2 ore body occurs in the lower part of the intrusion and is primarily composed of disseminated sulphide ore.

3 | PETROGRAPHY AND MINERALOGRAHY

The predominant rock types of the Jinchuan intrusion include dunite, lherzolite, olivine websterite, and minor plagioclase lherzolite (Tang

et al., 2006). In outcrops, dunite and lherzolite occur in the core, through olivine websterite to plagioclase lherzolite and websterite on the margin, without distinct boundaries between different lithofacies. The Jinchuan intrusion is characterized by olivine accumulations that comprise about 80 vol.% of the rocks. Alteration is pervasive in the Jinchuan intrusion. Olivine is partially replaced by serpentine plus secondary magnetite, and pyroxenes are partially replaced by chlorite plus tremolite or actinolite. The rocks between two ore bodies have been intensively affected by tectonic activities.

The main sulphide ore types in the Jinchuan deposit include disseminated, net-textured, Cu-rich sulphide ores, and massive sulphide ore.

1. Disseminated sulphide ores (Figure 3a) mainly occur in the western parts of Segments I, III, and IV, which are mainly hosted by lherzolite and olivine websterite. Bean-shaped and anhedral sulphide minerals are distributed in the interstitial voids of the olivine and pyroxene cumulates. The sulphide mineral assemblages comprise pyrrhotite, pentlandite, and chalcopyrite. Minor amounts of pyrite and chromite are also present. Pyrrhotite is the most abundant sulphide and makes up more than half of the total sulphide, with grain sizes varying from 0.1 to 3 mm. Pentlandite occurs as anhedral granular aggregates, "flame"

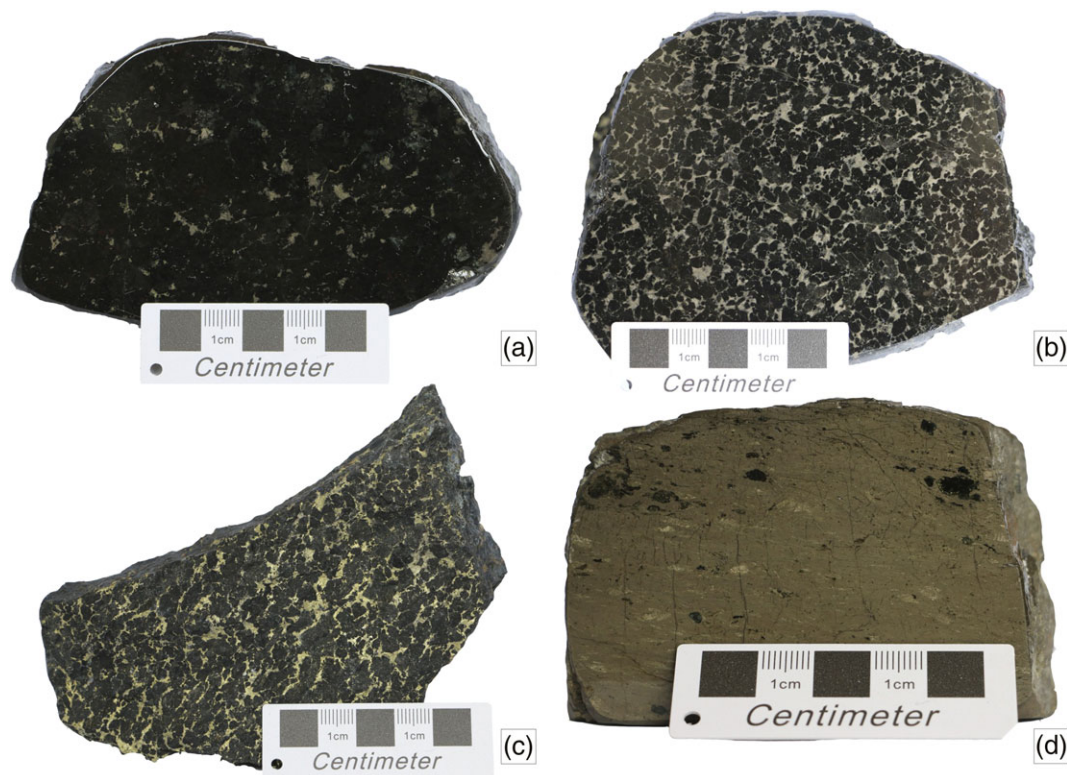


FIGURE 3 The main types of sulphide ore in the Jinchuan deposit. (a) Disseminated sulphide ore; (b) net-textured sulphide ore; (c) Cu-rich sulphide ore; (d) massive sulphide ore

exsolution, or euhedral crystals with variable grain sizes. Chalcopyrite predominantly occurs as anhedral crystals, and it either coexists with pyrrhotite or replaces pentlandite and pyrrhotite. Pyrite occurs rarely as thin vein.

2. Net-textured sulphide ore (Figure 3b) mainly occurs in the lower part of the intrusion in Segments I and II. The hosted rocks are dunite and lherzolite. The sulphide mineral assemblages of net-textured sulphide ores are similar to the disseminated sulphide ores, with sulphide minerals occurring in the interstitial voids of olivine cumulates. In addition, the disseminated and net-textured sulphide ores show gradual transitions, and minor abrupt variations can be ascribed to magma pulses.
3. Cu-rich sulphide ore (Figure 3c) mainly occurs between exploration Lines 6 to 8 in Segment I (Figure 2b) and extends along the late-stage fault F_{16} . The gradual transition from Cu-rich sulphide ores to net-textured sulphide ores and then to disseminated sulphide ores occurs along the underground section, without distinct boundaries between different ore types. The sulphide minerals comprise chalcopyrite, pyrrhotite, pentlandite, cubanite, and minor vallerite, violarite, pyrite, and platinum group minerals. Chalcopyrite predominantly occurs as anhedral crystals (0.1 to 0.5 mm) along the edges, fissures, and cleavages of pentlandite and pyrrhotite. The residual crystals of pentlandite and pyrrhotite are irregular and granular, with particle sizes of 0.1 to 2 mm.
4. Massive sulphide ore (Figure 3d) mainly occurs in the bottom of Segment II and the nearby surrounding rock, as well as a fraction between exploration Lines 15 and 24 in Segment I (Figure 2). The main minerals of the magmatic sulphide assemblage are

pyrrhotite, pentlandite, and chalcopyrite, with minor amounts of pyrite. Pyrrhotite makes up more than half of the total sulphides and occurs as anhedral crystal with occasional mechanical twins. Pentlandite occurs as subhedral to anhedral granular aggregates in proximity to pyrrhotite crystals (Craig, Vaughan, & Hagni, 1981; Gu & Vokes, 1996) or as exsolved "flame" with variable grain sizes. Chalcopyrite occurs as anhedral crystals (0.05 to 0.2 mm) and coexists with pyrrhotite.

4 | SAMPLING AND ANALYTICAL METHODS

In this paper, the selected samples are not affected by alteration, cracking, or late-stage deformation, including three massive sulphide ores and eight net-textured and disseminated sulphide ores, sampled from the No. 24 ore body of Segment I, exploration Lines 18–22, ~420 m below the surface, and five Cu-rich sulphide ores sampled from the No. 24 ore body of Segment I, exploration Lines 6–8, ~420 m below the surface. The sample locations are shown in Figure 2b.

4.1 | Mineralogical discrimination of pyrrhotite

The polished sections of different types of sulphide ores were examined using magnetic colloid techniques to identify the structure of pyrrhotite. Magnetic colloid contains abundant tiny magnetite grains following the preparation method given by Gu (1989). After applying the magnetic colloid, the reflectivity of monoclinic pyrrhotite (Mpo) significantly reduced relative to hexagonal pyrrhotite (Hpo) because Mpo is ferromagnetic whereas Hpo is paramagnetic, and the tiny

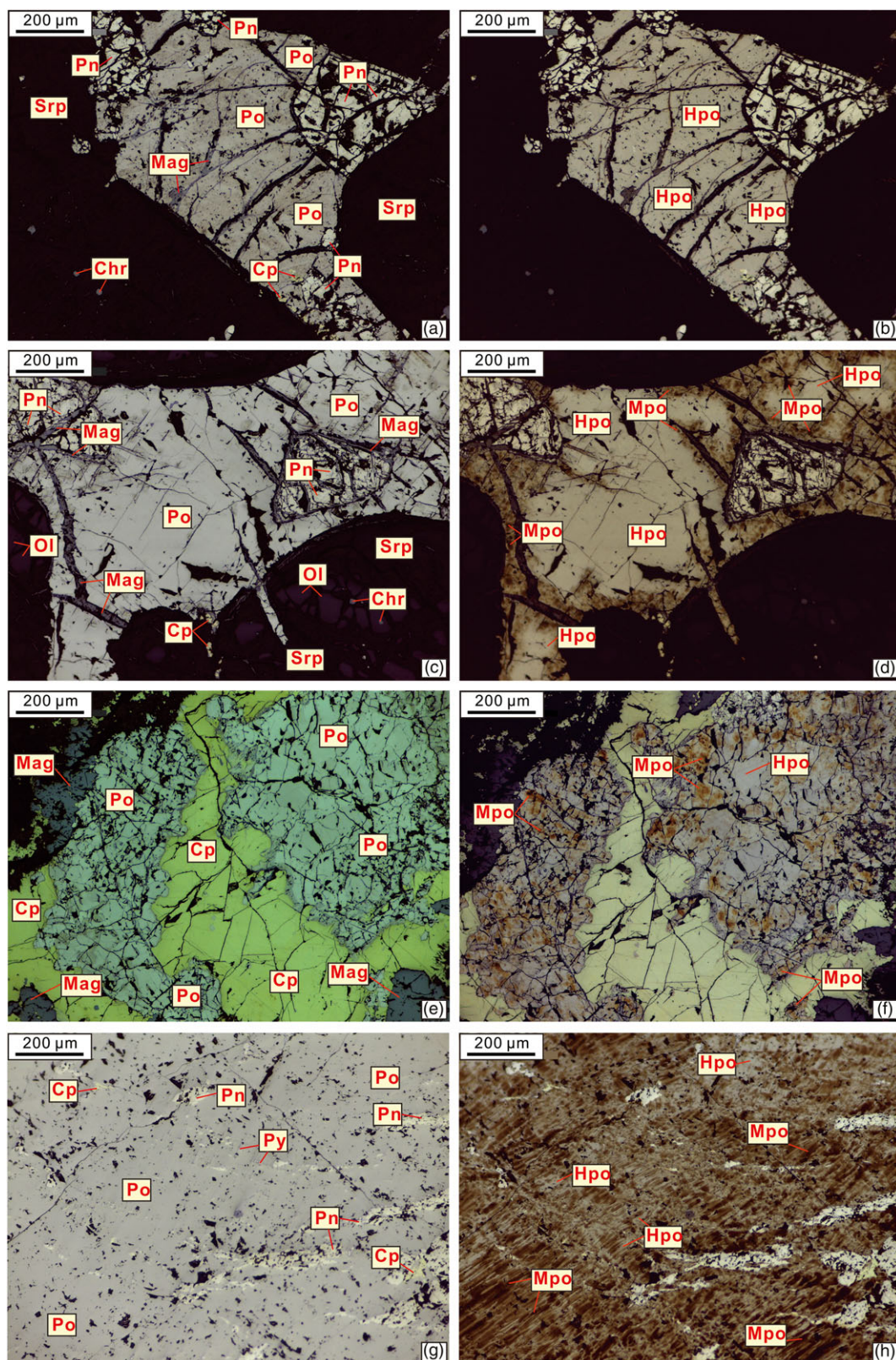


FIGURE 4 Photomicrographs of pyrrhotite showing the contrast before (a, c, e, g) and after (b, d, f, h) applying magnetic colloid. (a) Pyrrhotite and subhedral pentlandite in disseminated sulphide ores; (b) pure Hpo; (c) pyrrhotite in net-textured sulphide ores; (d) irregular intergrowths of Hpo and Mpo; (e) pyrrhotite in Cu-rich sulphide ores; (f) Mpo irregularly scattered in the Hpo; (g) pyrrhotite and granular polycrystalline veinlets in massive sulphide ores; (h) lamellar intergrowth of Hpo and Mpo. Chr = chrome spinel; Cp = chalcopyrite; Hpo = hexagonal pyrrhotite; Mag = magnetite; Mpo = monoclinic pyrrhotite; Ol = olivine; Pn = pentlandite; Po = pyrrhotite; Py = pyrite; Srp = serpentine

TABLE 1 Electron probe microanalysis data of pyrrhotite from the Jinchuan deposit

Ore types	Structure	As	S	Fe	Cu	Ni	Co	Total	Fe (at. %)
Disseminated sulphide ore	NC	0.05	38.56	60.28	–	0.02	–	98.91	47.31
		0.05	38.99	59.65	0.05	0.07	–	98.81	46.81
		–	38.47	60.10	0.06	0.13	–	98.76	47.36
		–	38.67	59.45	–	0.11	–	98.23	46.93
		0.06	38.27	60.37	0.05	0.04	–	98.79	47.56
		0.02	38.72	60.16	0.01	0.03	–	98.94	47.17
		0.08	38.51	59.59	–	0.14	–	98.33	47.11
		–	38.66	59.98	0.03	0.09	–	98.76	47.16
		–	38.71	59.80	0.02	0.16	–	98.68	47.08
		0.02	38.55	60.15	–	0.20	–	98.92	47.34
		–	38.58	59.55	–	0.09	–	98.22	47.02
		–	38.62	59.78	–	0.06	–	98.46	47.08
Net-textured sulphide ore	NC + dusty 4C	0.03	37.12	61.02	0.04	0.00	–	98.21	48.57
		–	37.39	62.16	–	–	–	99.55	48.84
		0.04	37.62	62.23	0.03	0.00	–	99.92	48.73
		–	37.23	61.27	–	0.00	–	98.50	48.59
		0.00	38.38	60.55	0.03	0.27	–	99.23	47.65
		–	38.89	60.20	0.00	0.22	–	99.31	47.15
		–	38.00	61.35	–	0.14	–	99.49	48.16
		–	37.29	62.17	–	0.02	–	99.48	48.92
		0.09	38.77	60.18	–	0.04	–	99.08	47.14
		0.02	39.05	60.04	–	0.01	–	99.11	46.89
		–	38.95	59.56	–	0.02	–	98.53	46.76
		0.07	38.36	60.03	0.02	0.03	–	98.51	47.35
		–	38.97	59.12	–	–	–	98.09	46.56
		–	38.90	59.77	0.01	0.00	–	98.68	46.88
		–	38.73	59.27	0.01	0.01	–	98.02	46.78
		–	38.67	59.52	0.05	0.07	–	98.31	46.96
		–	39.28	58.76	–	0.06	–	98.10	46.23
		0.01	38.91	59.78	0.02	0.07	–	98.79	46.91
		0.01	38.83	59.86	–	0.01	–	98.71	46.96
		0.05	38.91	59.82	–	0.04	–	98.82	46.90
Cu-rich sulphide ore-1	NC + dusty 4C	0.03	39.21	59.52	0.00	0.00	0.13	98.89	46.63
		0.05	38.79	60.04	0.02	0.04	0.11	99.05	47.12
		0.00	39.48	59.74	0.05	0.00	0.07	99.34	46.54
		0.00	39.39	59.37	0.03	0.02	0.11	98.92	46.46
		0.00	38.99	59.03	0.09	0.31	0.12	98.53	46.71
		0.00	39.17	59.29	0.06	0.18	0.10	98.79	46.63
		0.00	39.34	59.36	0.03	0.04	0.11	98.88	46.49
		0.00	39.03	59.36	0.02	0.04	0.12	98.57	46.69
		0.00	39.20	59.43	0.03	0.07	0.12	98.85	46.62
		0.05	39.48	59.23	0.03	0.05	0.13	98.98	46.36
		0.02	39.14	59.34	0.02	0.06	0.08	98.65	46.60
		0.00	39.43	59.24	0.01	0.17	0.10	98.94	46.42
		0.00	39.14	59.22	0.05	0.20	0.10	98.70	46.63
		0.00	39.06	59.19	0.05	0.32	0.09	98.72	46.71
		0.00	39.55	59.03	0.05	0.24	0.11	98.97	46.31
		0.00	39.60	59.30	0.03	0.22	0.09	99.24	46.37
		0.00	39.20	59.19	0.05	0.31	0.08	98.83	46.61
		0.00	39.47	59.28	0.05	0.28	0.12	99.20	46.49
Cu-rich sulphide ore-2	NC + dusty 4C	0.01	39.31	59.55	0.05	0.08	0.10	99.10	46.61
		0.04	39.04	59.64	0.08	0.00	0.10	98.90	46.80
		0.00	39.42	59.68	0.07	0.04	0.10	99.31	46.59
		0.03	39.20	59.35	0.05	0.11	0.09	98.83	46.60
		0.00	39.21	59.20	0.01	0.11	0.12	98.64	46.53
		0.00	38.81	59.12	0.06	0.12	0.12	98.23	46.78
		0.00	39.38	59.61	0.04	0.15	0.10	99.28	46.61
		0.04	39.22	59.52	0.05	0.08	0.10	99.01	46.66
		0.00	39.42	58.77	0.14	0.09	0.11	98.53	46.25
		0.00	39.01	59.29	0.08	0.08	0.12	98.58	46.71
		0.03	39.01	59.96	0.03	0.00	0.09	99.11	46.93
		0.00	38.83	59.88	0.02	0.05	0.12	98.90	47.03
		0.01	38.64	59.79	0.00	0.00	0.10	98.54	47.09
		0.01	38.71	59.93	0.06	0.00	0.12	98.82	47.13
		0.00	39.08	60.25	0.03	0.02	0.13	99.51	47.02
		0.00	38.90	59.81	0.03	0.00	0.12	98.86	46.95
		0.02	38.84	59.90	0.01	0.01	0.12	98.89	47.02
		0.00	38.41	59.44	0.03	0.06	0.10	98.04	47.13
Massive sulphide ore-1	NC + lamella intergrowth 4C	–	40.18	57.73	–	0.39	–	98.30	45.37
		–	40.26	57.40	0.02	0.51	–	98.19	45.23
		0.03	40.54	57.41	–	0.36	–	98.34	45.00

(Continues)

TABLE 1 (Continued)

Ore types	Structure	As	S	Fe	Cu	Ni	Co	Total	Fe (at. %)
		–	40.38	58.55	0.00	0.38	–	99.32	45.59
		0.01	40.75	57.74	–	0.36	–	98.85	45.01
		–	40.61	57.62	0.03	0.49	–	98.75	45.11
		–	40.21	57.34	–	0.50	–	98.05	45.23
		–	40.53	57.04	–	0.60	–	98.17	44.94
		–	40.62	58.15	–	0.40	–	99.17	45.28
		–	40.63	58.09	–	0.66	–	99.38	45.35
		–	40.02	57.63	–	0.47	–	98.12	45.45
		0.09	40.51	57.07	–	0.52	–	98.19	44.93
		0.09	40.40	58.25	0.03	0.38	–	99.15	45.46
		–	40.56	57.84	0.05	0.42	–	98.87	45.21
		–	40.17	57.73	–	0.45	–	98.35	45.40
		–	40.36	57.23	–	0.44	–	98.03	45.06
		–	40.63	57.57	–	0.45	–	98.65	45.04
		0.00	40.13	57.61	–	0.59	–	98.33	45.43
		0.04	40.31	57.85	0.04	0.64	–	98.88	45.45
		0.07	40.52	58.35	–	0.46	–	99.41	45.45
		–	40.45	57.18	–	0.56	–	98.19	45.03
		0.07	40.07	58.40	0.02	0.40	–	98.96	45.73
		–	40.60	57.93	–	0.37	–	98.90	45.18
		0.07	40.65	57.62	0.01	0.67	–	99.02	45.15
		0.02	40.36	57.75	0.03	0.68	–	98.83	45.39
		0.03	40.24	57.41	–	0.85	–	98.52	45.38
		–	40.51	57.93	–	0.51	–	98.95	45.30
		–	40.55	57.42	0.01	0.52	–	98.50	45.06
Massive sulphide ore-2	NC + lamella intergrowth 4C	0.05	39.15	58.66	–	0.69	–	98.55	46.53
		–	39.27	58.83	–	0.53	–	98.63	46.46
		0.02	38.95	59.14	–	0.41	–	98.52	46.74
		–	39.03	58.55	–	0.43	0.03	98.04	46.46
		0.01	39.24	59.27	0.01	0.51	–	99.04	46.65
		0.01	39.02	58.70	0.06	0.41	–	98.19	46.53
		0.04	39.49	58.46	–	0.52	–	98.51	46.16
		0.06	39.15	58.72	–	0.45	0.02	98.40	46.46
		0.03	39.06	59.43	0.05	0.48	–	99.05	46.84
		0.03	39.29	58.44	–	0.45	–	98.22	46.25
		–	39.42	58.69	–	0.39	–	98.50	46.25
		0.02	39.24	58.74	–	0.46	–	98.45	46.41
		–	39.29	58.82	–	0.56	–	98.67	46.45
		0.05	39.46	58.55	0.05	0.48	0.01	98.60	46.22
		0.05	38.94	59.08	–	0.40	–	98.48	46.72
		0.01	39.13	59.42	–	0.42	–	98.97	46.75
		–	39.26	58.62	0.03	0.43	–	98.34	46.34
		–	39.19	58.65	0.07	0.65	–	98.55	46.50
		0.09	39.37	58.15	–	0.53	–	98.14	46.11
		–	39.72	58.77	–	0.43	–	98.92	46.11
		–	39.62	59.17	0.00	0.41	–	99.20	46.33
		0.04	39.58	58.91	0.01	0.33	–	98.87	46.22
		0.07	39.13	58.80	0.00	0.46	–	98.47	46.50

magnetites will adhere to the Mpo. The quantity and distribution pattern of Mpo can be easily distinguished. The results of this method are completely consistent with those of traditional methods (e.g., chromic acid etching), and the processes used in this method are more simple and convenient than X-ray powder diffraction analysis (Gu, Yin, & Vokes, 1995).

4.2 | Electron probe microanalysis of pyrrhotite

We obtained the composition of pyrrhotite using electron probe microanalysis and estimated the structural type of pyrrhotite based on the percentage of atomic iron (at. %). Pyrrhotite compositions were determined by wavelength dispersive X-ray emission microanalysis using a JXA-8100 electron microprobe at the Key Laboratory of Western China's Mineral Resources and Geological Engineering, Ministry of Education, China. The analytical conditions for major elements were an accelerating voltage of 15 kV, a beam current of 20 nA, a beam size of

1 μm , and a peak counting time of 20 s. The analytical reproducibility was within $\pm 2\%$ relative standard deviation.

5 | ANALYTICAL RESULTS

5.1 | Structure of pyrrhotite

The magnetic colloid experiment revealed that the structure of pyrrhotite from the Jinchuan deposit can be classified into three types:

1. Pure Hpo occurs in the disseminated and net-textured sulphide ores. Pyrrhotite is a simple Hpo (Figure 4a,b), lacking of Mpo.
2. Irregular intergrowth pyrrhotite occurs in disseminated, net-textured, and Cu-rich sulphide ores. The dusty, stellate, and patchy Mpo occurs along the fissures and grain boundaries of Hpo or at the contact boundaries of silicate minerals (Figure 4c,d). Mpo and Hpo have variable ratios from 0:1 to 0.2:1. In the Cu-rich sulphide

ores, the Mpo is irregularly scattered within the Hpo, and the distribution is not controlled by the sulphide boundaries (Figure 4e,f).

3. Lamellae intergrowth pyrrhotite occurs in the massive sulphide ores. The pyrrhotite crystals are found as the intergrowth of Hpo and Mpo, and the Mpo occurs as roughly parallel lamellae within the matrix of Hpo (Figure 4g,h). The sizes of the Mpo lamellae range from 10 to 20 μm in width and 2 to 30 μm in interval. Mpo and Hpo have variable ratios from 0.2:1 to 2:1. The Mpo lamellae intersect at low angles or are parallel to the major axis of the exsolved "flames" of pentlandite and the fine-grained pentlandite (Figure 4g,h).

In conclusion, the characteristics of sulphide intergrowth and the ratios of Hpo and Mpo in net-textured, disseminated, and Cu-rich sulphide ores are very similar, yet distinctly different from those associated with massive sulphide ores.

5.2 | Mineral chemistry of pyrrhotite

One hundred and forty-three analyses of pyrrhotite composition by electron probe microanalysis (Table 1) and magnetic colloid techniques show continuous variation of Fe and S concentrations from Hpo to Mpo (Figure 5a). The Hpo has the highest Fe concentration and the lowest S concentration, ranging from 59.32% to 62.50% and from 37.49% to 40.05%, respectively. The Fe and S concentrations of Hpo + dusty Mpo are 58.77% to 60.94% and 38.41% to 40.04%, respectively. In contrast, the Hpo + lamellae Mpo have the lowest Fe concentration and the highest S concentration from 58.11% to 60.04% and from 39.44% to 41.31%, respectively. Hpo has a larger variation of Ni content, yet the Hpo + lamellae Mpo display a smaller range of variation (Figure 5b). The Ni and S concentrations show a positive correlation (Figure 5b), but the Ni and Fe concentrations show a negative correlation (Figure 5c).

When the activity of FeO (α_{FeO}) is constant, the ratio of M/S (in which M includes Fe, Ni, Cu, and Co atoms) decreases with increasing of oxygen fugacity during crystallization of monosulphide solid solution (MSS; Figure 5d; Naldrett et al., 2000). The M/S values of disseminated, net-textured, and Cu-rich sulphide ores are 0.88 to 0.91 (with an average of 0.89), 0.86 to 0.96 (with an average of 0.90), and 0.86 to 0.89 (with an average of 0.88), respectively. The M/S values of pyrrhotite in massive sulphide ores are 0.82 to 0.88 (with an average of 0.85). Therefore, the M/S values of pyrrhotite in the net-textured, disseminated, and Cu-rich sulphide ores are generally higher than those of massive sulphide ores.

Furthermore, we selected two areas with a diameter of 1 mm in the pyrrhotite from two samples of massive sulphide ores and analysed the composition of pyrrhotite every $\sim 50 \mu\text{m}$ along the vertical direction of the lamellae Mpo (Figure 6). The atomic percent of Fe (at. %) shows continuous fluctuations, which could be ascribed to analytical error. The atomic percent of Fe (at. %) of pyrrhotite-dominated samples in massive sulphide ores (Figure 6b,d) is higher than that of samples with both pyrrhotite and pentlandite (Figure 6a,c), suggesting that pyrrhotite crystallized earlier than pentlandite.

6 | DISCUSSION

6.1 | Genesis of different structures in pyrrhotite

The pyrrhotite in the Jinchuan deposit was formed by magmatic crystallization, and the structures of pyrrhotite are similar to that from some magmatic Ni-Cu sulphide deposits in Xinjiang, NW China (Chen, 1995; Ding et al., 2007; Hao, 1986), and the Voisey's Bay Cu-Ni sulphide deposit in Labrador, Canada, which represents the most reduced magmatic sulphide deposit known worldwide (Naldrett et al., 2000).

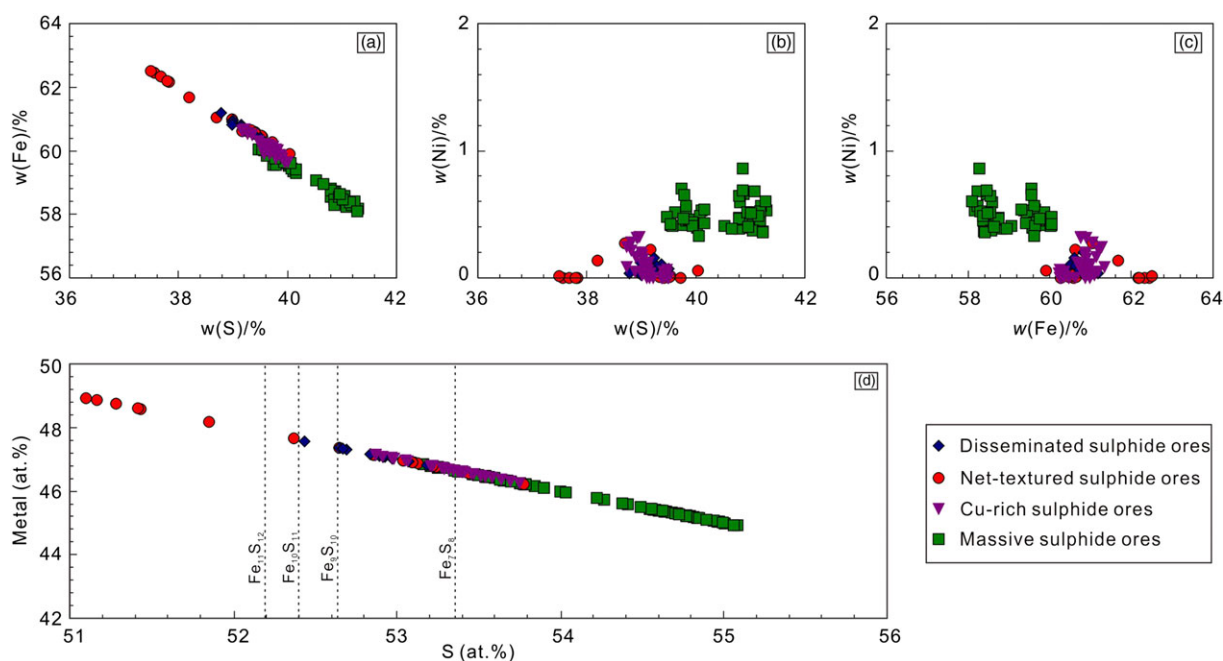


FIGURE 5 Mineral compositions of pyrrhotite from different sulphide ores of the Jinchuan deposit

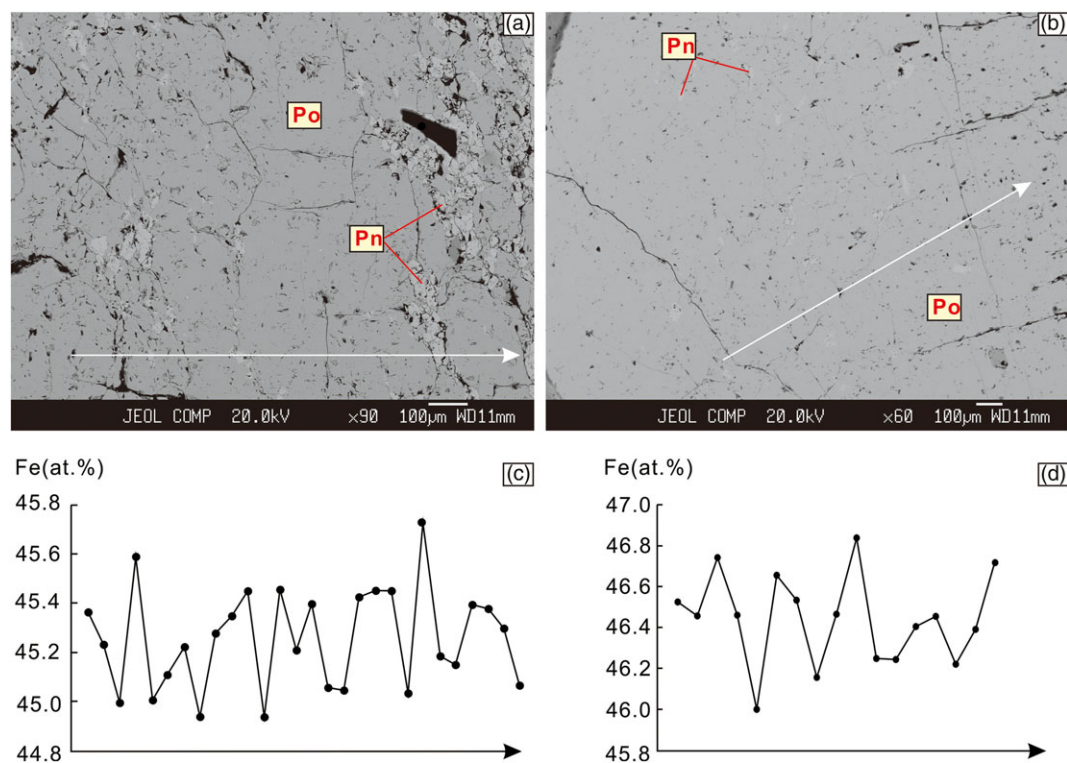


FIGURE 6 Back-scattered electron (BSE) images and compositional profiles of pyrrhotite in massive sulphide ores. (a) Pyrrhotite and polycrystalline pentlandite in massive sulphide ore; (b) pyrrhotite and granular veinlets of pentlandite in massive sulphide ore; (c) compositional profiles of Fe (at. %) in pyrrhotite from Figure 6a; (d) compositional profiles of Fe (at. %) in pyrrhotite from Figure 6b; Pn = pentlandite; Po = pyrrhotite

After the classification and calculating of the electron probe analytical data, a frequency distribution histogram was plotted with metal atoms (Fe, Ni, Cu, and Co) on the abscissa (Figure 7). Naldrett et al. (2000) sorted the structures of pyrrhotite into NC intergrown with 2C, and pure NC in Voisey's Bay deposit. The pyrrhotites in disseminated ores in the Jinchuan deposit fall into the 4C + NC and NC phase areas; the pyrrhotites in net-textured sulphide ores fall into the 4C, 4C + NC, NC, and 2C + NC phase areas; the pyrrhotites in Cu-rich sulphide ores fall into the 4C + NC and 4C phase areas; and the pyrrhotites in massive sulphide ores fall into the 4C + Py and 4C phase areas. In all, the pyrrhotites in disseminated, net-textured, and Cu-rich sulphide ores are mostly Hpo, but the pyrrhotites in massive sulphide ores are Hpo + lamellar Mpo, consistent with the results of magnetic colloid experiments (Figure 7).

It is generally believed that Hpo and Mpo could occur as two different intergrowths (Gu & Vokes, 1996; Gu, Xiao, Ni, Wu, & Mao, 2001; Gu, Zhen, & Lu, 1988; Gu, Zheng, Tang, & Wu, 2006).

6.1.1 | Lamellae intergrowth formed by exsolution

Hpo crystallized at high temperature and had sufficient time to form pyrite exsolution (Arnold, 1962; Yund & Hall, 1970), whereas pyrite porphyroblasts formed during the slow cooling process (Gu et al., 2001; Gu & Vokes, 1996). When excess sulphur in the MSS combined with Fe to form pyrite, a combination of Hpo and pyrite will be formed. However, if the temperature decreases rapidly, the excess sulphur will have no enough time to exsolve as pyrite. As the temperature

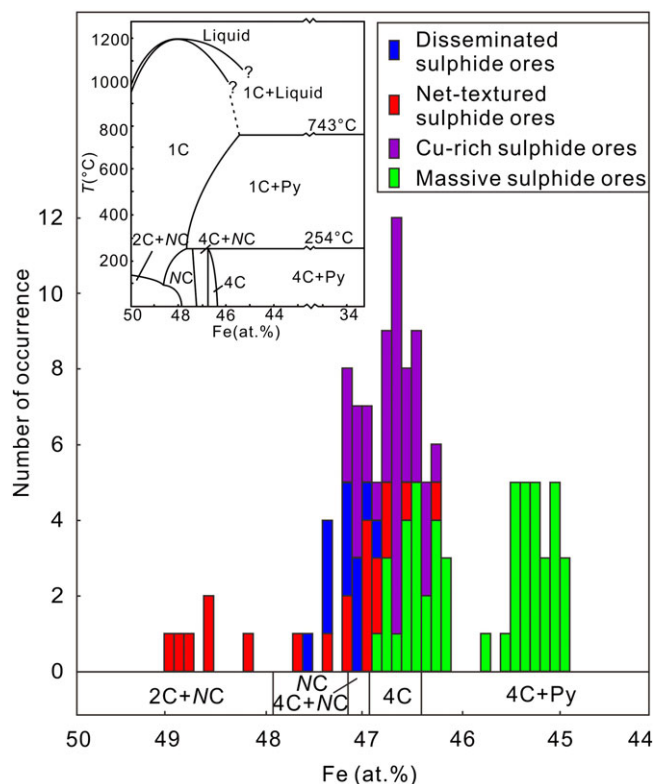


FIGURE 7 Histogram of the variations in atomic metal content following pyrrhotite phase equilibrium diagram (after Arnold, 1962; Desborough & Carpenter, 1965; Kissin & Scott, 1982)

decreased to 254 °C, the Mpo would be exsolved (Kissin & Scott, 1982), resulting in the intergrowth of lamellar Mpo in the Hpo matrix (Gu et al., 2001; Gu & Vokes, 1996).

6.1.2 | Irregular intergrowth resulting from metasomatism

The origin of irregular intergrowths can be explained by hydrothermal alteration. In this scenario, the irregular intergrowths that formed on both sides of the fissures of the Hpo and the edges of the particles (Gu et al., 1988, 2006) could have resulted from the interaction of Hpo and sulphur-rich and/or oxygen-rich hydrothermal fluids.

In the Jinchuan deposit, two distinct intergrowths between Mpo and Hpo were formed at the same time. The Hpo + lamellar Mpo in massive sulphide ores formed by rapid cooling. When the temperature decreased to 254 °C, the pyrrhotites entered the 4C + NC stability phase areas. The pure Hpo in the net-textured and disseminated sulphide ores was formed during the slow cooling process, whereas the Hpo + dusty Mpo were resulted from the interactions of Hpo and sulphur- and/or oxygen-rich hydrothermal fluids (Figure 8).

6.2 | Genesis of different types of sulphide ore

6.2.1 | Genesis of the disseminated and net-textured sulphide ores

The disseminated and net-textured sulphide ores are ubiquitous in the Jinchuan deposit, and they were proposed to have formed by in situ sulphide segregation (Tang & Li, 1995). The crystal structure and mineral chemistry of pyrrhotite from the two types of sulphide ore are similar. Pure Hpo could have formed during slow cooling after crystallization without the effect of hydrothermal fluids.

In addition, there are two other possible mechanisms for the formation of pure Hpo: (a) low-S Hpo crystallizing from low-S magma, limiting the exsolution of Mpo during cooling, and (b) the homogenization of exsolved Mpo in the Hpo due to late-stage metamorphism (Gu & Vokes, 1996). In the Jinchuan deposit, there are no deformation or re-equilibrium structures due to annealing, suggesting that late-stage metamorphism was unrelated to the formation of Hpo. The absence of porphyroblastic pyrite in disseminated and net-textured sulphide ores, as well as the scattering of high M/S pyrrhotite in 4C + NC and its adjoining phases (Figure 7), indicates a very low-S magma system. Thus, the pure Hpo in disseminated and net-textured sulphide ores appears to be the product of slow cooling following high-temperature crystallization in a low-S magma system (Figure 8).

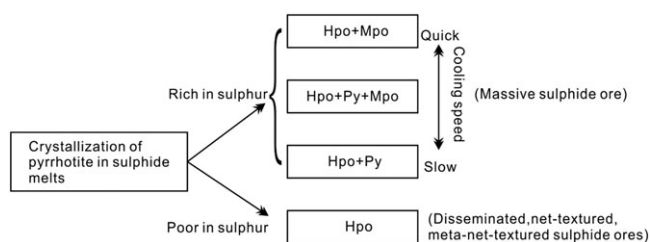


FIGURE 8 Crystallization process of pyrrhotite in sulphide melts of the Jinchuan deposit

6.2.2 | Genesis of the Cu-rich sulphide ores

Cu-rich sulphide ores are characterized by a relatively high proportion of chalcopyrite and high Pt and Pd concentrations, but low IPGE (Os, Ir, and Ru) concentrations (Chen et al., 2013), consistent with the characteristics of residual sulphide melt mineralization after MSS separation (Barnes & Lightfoot, 2005). Some previous studies have proposed three genetic models for the Cu-rich sulphide ores, including hydrothermal superimposition (Tang & Li, 1995), late-magmatic products (Tian, Bao, Tang, & Wang, 2009), and late-stage magma injection (Su, Tang, & Zhou, 2010). The sulphide association dominated by chalcopyrite, pentlandite, and pyrrhotite is inconsistent with later hydrothermal superimposition (Franchuk, Lightfoot, & Kontak, 2016; Lightfoot, Stewart, Gribbin, & Mooney, 2017). On the contrary, the mineral structure and chemistry of pyrrhotite in the Cu-rich sulphide ores are consistent with those of the disseminated and net-textured sulphide ores, and they fall into the 4C + NC and 4C phase areas, indicating that they crystallized during slow cooling after high-temperature crystallization of a low-S magma, similar to those from the disseminated and net-textured sulphide ores. Therefore, the genesis of Cu-rich sulphide ores appears to be associated with late magmatic mineralization. Sulphide melts have experienced fractional crystallization of MSS, with the residual Cu–Pt–Pd-rich sulphide melts being locally concentrated and mineralized.

6.2.3 | Genesis of the massive sulphide ores

The massive sulphide ores often invaded the disseminated, net-textured sulphide ores or surrounding rocks. Previous studies have suggested that the massive sulphide ores were formed by the segregation of sulphide melts in a deep chamber and the subsequent emplacement of multiple injections (Jiao et al., 2012; Su et al., 2010; Tang, 1996; Wang, Mao, He, Wang, & Tang, 2004; Xu et al., 2012; Yan, Su, & Jiao, 2005). The sulphide melt was formed by the separation and concentration of immiscible sulphide melts within a magma chamber and then injected as small volume into the consolidated net-textured and disseminated sulphide ore bodies or surrounding rocks, resulting in the rapid decrease of temperature of the sulphide melts. Subsequently, massive sulphide ores containing lamellae intergrowth structure of the pyrrhotite were formed from the sulphide melt due to rapid drop of the temperature. The Hpo, which crystallized at high temperature, had insufficient time to form pyrite porphyroblasts in a high-S magma system (Figure 8). Mpo would be exsolved from Hpo when the temperature decreased to 254 °C, resulting in the intergrowth of lamellar Mpo in Hpo matrix (Figure 8). Overall, the absence of pyrite porphyroblasts in lamellae intergrowth of pyrrhotite in massive sulphide ores suggests a rapid reduction of temperature.

7 | CONCLUSIONS

1. Disseminated and net-textured sulphide ores in the Jinchuan deposit were formed by in situ sulphide segregation. Pyrrhotite from these two ore types formed in a slow cooling environment after crystallization in a high-temperature and low-S magma system.

2. Cu-rich sulphide ores are the residual sulphide melts after the fractional crystallization of MSS. Pyrrhotite in the Cu-rich sulphide ores were produced by slow cooling process after high-temperature crystallization in a low-S magma system, similar to those from the disseminated and net-textured sulphide ores.
3. Massive sulphide ores were formed by injection or flow of the immiscible sulphide melts and experienced a rapid reduction of temperature in a high-S magma system.

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